

22GE004 BASICS OF ELECTRONICS

Unit I & Electrical and Electromagnetic Signal Transmission Signal transmission

Electrical Signal Transmission

Any signal, light/sound/video that is sent over a wire (conductor) by drifting of electrons in it (current) caused variation of electrical voltages at either end. They need a conductor to propagate. eg: Electricity coming to your house, or the music from your cell phone, through the earphone wires.

5.1 Transmission of electrical signals on wires

In the discussions of newsgathering in the Taylor and Higgins papers, you saw the significance of the development of systems that allowed long-distance transmission of electronic signals. Initially transmission used metallic wires (remember Taylor's reference to the importance of the 'lines infrastructure' and his mention of the 'wire picture') and later wireless transmission (terrestrial and satellite microwave) became important. In this section I shall look at some aspects of the transmission of digital signals, starting with a close look at the transmission of electrical signals on wires.

Using a wire to transmit a signal is simple in principle: you operate a switch at one place and observe the effect somewhere else. **In Figure 1** I have shown this as a light coming on at a remote location. Notice in this diagram the standard symbol for a battery consisting of two parallel lines, one shorter than the other, and the symbol for a light bulb which is a circle with a cross in it. The standard symbol for a switch that can be 'open' or 'closed' consists of two dots and a line which either connects the dots (when the switch is closed as in **Figure 1 b**) or misses one of the dots (when the switch is open, as in **Figure 1 a**). To switch the light on you close the switch so that there is an electric circuit from the battery to the light bulb and back again. To switch the light off you open the switch to 'break' the circuit.

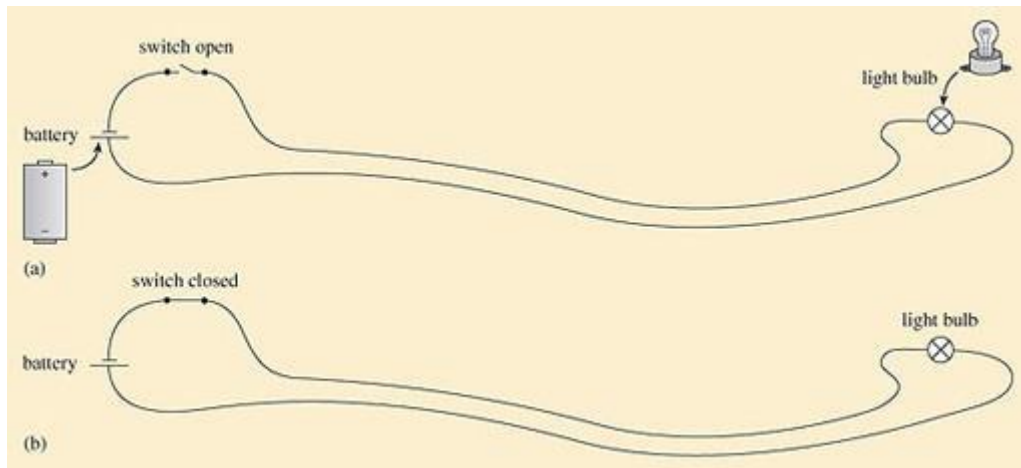


Figure 1 Switching a light on at a remote location: (a) light off; (b) light on

When you switch a light on, the light appears to come on immediately. There does not appear to be any delay between operating the switch and the effect at the light (although, depending on the type of light there might be a delay before it comes on fully – this is especially noticeable with fluorescent light tubes). In reality there is a delay – it is just very short indeed.

To get a better insight into what is happening, imagine measuring the voltage between the wires. This can be done with something called a **voltmeter** (**Figure 2**). A voltmeter has two wires and a display. When you touch the ends of the wires to the terminals of a power source like a battery, the display on the meter tells you what the voltage is between the terminals.



Figure 2 Voltmeter

Imagine using the voltmeter to measure the voltage between the two wires at some point between the switch and the light, as in **Figure 3**. When the switch is open (off) the reading on the meter will be zero. When the switch is closed (on), the reading will (ideally) be equal to the voltage of the battery – which I shall assume is 1.2 volts. Now imagine having the voltmeter touching the wires while the switch is changed from open to closed. In this case you will see the voltage change from 0 to 1.2 V.

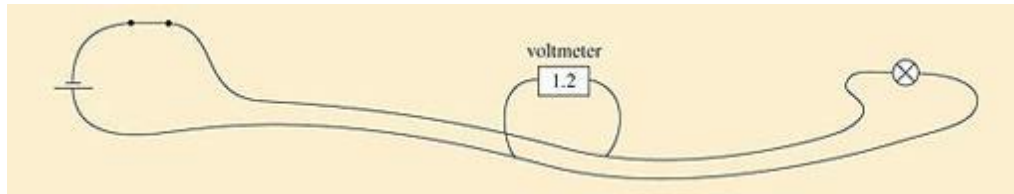


Figure 3 Measuring the voltage across a pair of wires

Now, if the voltmeter is touching the wires right next to the switch, you would see the voltage rise from 0 to 1.2 V at the same instant as the switch is closed. If, on the other hand, the voltmeter is touching the wires further away from the switch there will be a delay between the switch closing and the voltage rising. We can display this by plotting graphs as shown in **Figure 4**.

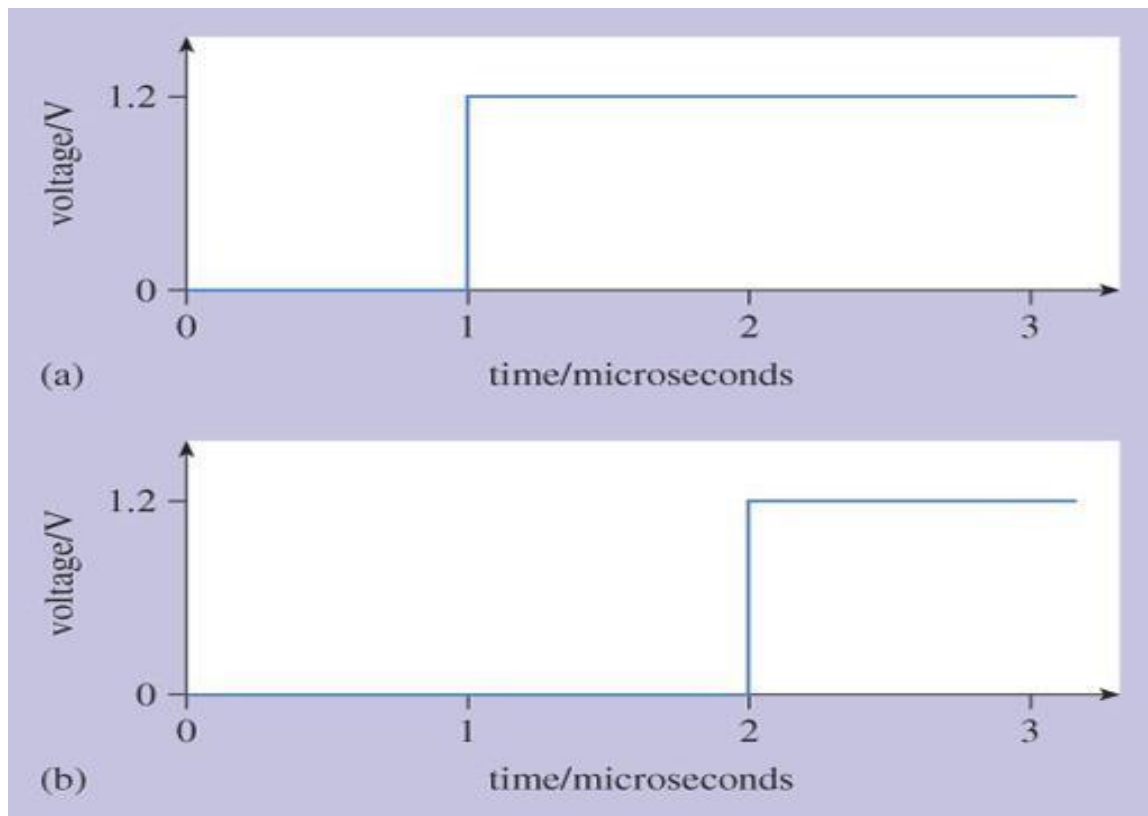


Figure 4 The voltage across wires when a switch is closed: (a) voltage across the wires as measured at the switch; (b) voltage across the wires as measured 200 m away from the switch

These graphs show how the reading on the voltmeter changes with time. Along the horizontal axis from left to right corresponds to time passing, and up the vertical axis corresponds to increasing voltage, as measured by the voltmeter.

By convention, the axis that goes across the page is the 'horizontal' axis, and the axis that goes up and down the page is the 'vertical' axis. Sometimes they are called the x and y axes, for horizontal and vertical respectively.

The time axis is labelled in units of microseconds, where one microsecond is one-millionth of a second. Notice also that the time axis is relative to some time origin which is labelled 0. The actual time corresponding to 0 as shown on the axis might have been, say, Thursday 20 May 2004, 2.17 pm and 35.031233 seconds, but labelling the axis with that level of detail would be confusing and irrelevant.

Figure 4(a) shows what the voltmeter would do when connected to the wires next to the switch, while **Figure 4(b)** shows what it would do when connected to the wires 200 m along towards the light bulb. The switch was closed at time 1, on this scale, so the voltage measured next to the switch rises at time 1.

There is a delay before the voltage rises at the voltmeter when it is 200 m along the wire.

5.2 Other transmission media

Wires are still used to carry electrical signals over short distances. At the time of writing, for example, most connections between telephones in private houses and the local telephone exchange still use wires. The telephone networks within office buildings are mostly connected with wires, and so are many computer networks (local area networks, LANs) within single buildings. However, all longer-distance communication, between towns, cities or countries, uses either optical fibre or microwave systems. Increasingly, even shorter distances use either optical fibres or wireless links of one sort or another.

Many of the concepts described in the previous section for signal transmission on electrical wires also apply to wireless systems (such as microwave) or optical fibres. Pulses are still attenuated and distorted, for example. It is worth, however, briefly looking at some of the characteristics particular to these other transmission media.

5.2.1 Microwave

You saw the importance of microwave transmission for newsgathering in the Higgins extract. The term 'microwave' identifies a particular range of frequencies used for radio communications. The range of frequencies that are referred to as 'microwave' is not exactly defined (or, rather, slightly different ranges are used in different contexts), but roughly speaking it is from about 200 MHz to 50 GHz. [Remember that MHz stands for megahertz, which is 1,000,000 Hz (10^6 Hz) and GHz is gigahertz, which is 1,000,000,000 Hz (10^9 Hz).]

It is possible to transmit digital signals over microwave by using pulses of microwave power to represent 1s, and the absence of microwave power to represent 0s. This type of transmission is known as on-off keying. In practice this is not generally the best way to use microwave for transmission, and more sophisticated ways of putting the data on the radio signal (different **modulation schemes**) are normally used.

Regeneration is used in microwave transmission systems, but surprisingly long distances are possible without regeneration. In satellite communications, the satellite performs regeneration (as well as some other functions), but that still means that the signal has to travel the distance from the ground to the satellite in one go (and the same distance back). For geostationary satellites this is 36,000 km each way.

Even more remarkable is the microwave transmission that was used to send data back from the Cassini–Huygens space exploration mission to Saturn and its moon, Titan, in 2004/05. The distance between the Earth and Saturn was at that time 1,517,000,000 km! The reduction in the signal strength over that distance is

very great, so the power of the signal received back on the Earth is small. In practice this means that the data rate (bandwidth) is small, because there is a trade-off between signal power and data rate. For high data rates higher power is needed. If the power is low, only low data rates are possible. (It is the same if you are having difficulty hearing someone speaking, when you might ask them to speak more slowly.

5.2.2 Optical fibre

In all developed countries, long-distance communication links (which used to be called 'trunks', by analogy to 'trunk road') nearly always use optical fibre. It is only where the terrain makes it difficult to lay a cable (such as in mountains or, sometimes, between islands) or when a new link is needed quickly and there isn't time to lay a cable that microwave links are used instead.

An optical fibre is a strand of glass or plastic, not much thicker than a human hair, which guides light from one end to the other (**Figure 5**). The guidance comes about because of an effect known as total internal reflection. This means that light shone in one end of the fibre doesn't come out from the sides of the fibre even if the fibre is bent around corners. The light just travels inside the fibre until it comes to the far end. Because the fibre is so thin it is flexible and, from the outside of a cable with protective plastic coverings, looks and feels much the same as an electrical wire. Signals are conveyed by changing the 'brightness' of the light injected into the fibre and measuring it at the far end. Bits are sent by 'on-off keying': 1s are represented by light on and 0s by light off.



Figure 5 Optical fibre of the type used for communications. The bare glass fibre is vulnerable to scratches and will break if bent too tightly, so the cable shown has several layers of protective coverings, shown here stripped off layer by layer.

The attraction of optical fibre is that it can be used for very high data rates over long distances. It is this combination – high data rates and long distances – that distinguishes it from wires.

When electrical signals are transmitted over wires the attenuation increases with increasing data rate, so the higher the data rate, the greater the attenuation and therefore the shorter the distance that can be used. So, for example, although signals at 1 gigabit/s can be carried around on wires inside a computer, it is difficult to transmit electrical signals even a few metres at data rates that high. With optical fibre the attenuation is not dependent on the data rate, and the attenuation is anyway very low, so that signals at tens or even hundreds of gigabit/s can be sent for tens of kilometres with the right sort of fibre.

There are other factors to consider, including, as mentioned above, that higher data rates require higher power (which is the case whatever the medium used to carry the signal). Nevertheless, even though optical fibre is generally more expensive to use than wires, optical fibre is the transmission medium to use for high data rates over long distances.

5.3 Signal speeds, propagation times and distance: the formula triangle

When signals travel along a wire or optical fibre, or through space, the relationship between the speed, propagation time and distance can be written in three ways, depending upon which one you want to calculate.

If you know the speed and the propagation time and want to know how far the signal will travel, you use:

$$\text{distance} = \text{propagation time} \times \text{speed}$$

So we have

$$d = t v$$

Technically the value for the velocity of something includes both its speed and the direction it is moving in. For our purposes there is no need to distinguish between speed and velocity. I shall just use the word speed and the symbol v .

If you know the distance travelled and speed but want to calculate the propagation time you use:

$$\text{propagation time} = \frac{\text{distance}}{\text{speed}}$$

which can be written:

$$t = \frac{d}{v}$$

Finally, if you know the distance travelled and the propagation time, and want to calculate the speed, you use:

$$\text{speed} = \frac{\text{distance}}{\text{propagation time}}$$

Or

$$v = \frac{d}{t}$$

These are of course quite general relationships between distance travelled, time taken and speed. We used the same relationship when doing the calculations for autofocus in a camcorder, and it applies equally for the journey time driving along a motorway or cycling to work, assuming constant speed (or using average speed in the calculation). The relationship between these three terms is displayed on a formula triangle as shown in **Figure 6**. You can swap the 'time' t and 'speed' (v) in the lower corners – it doesn't matter which way round these are.

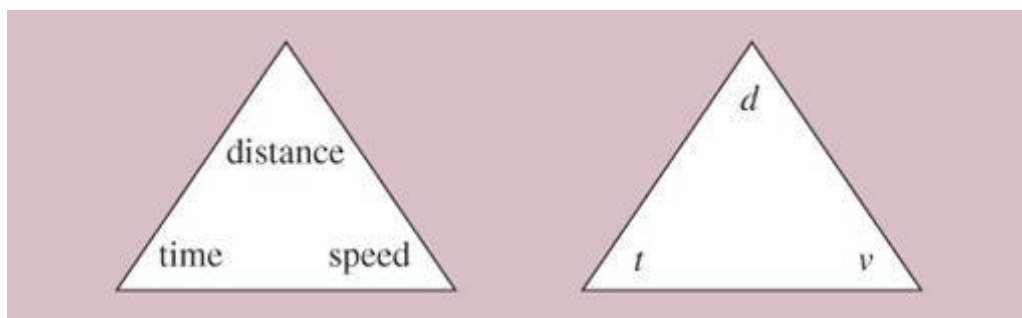


Figure 6 The formula triangle

Earlier I explained how you draw the formula triangle given one of the equations, but what is useful about the triangle is that you can get back to any of the equations from the triangle.

You do this by covering the quantity you want to calculate, and looking at the position of the other two (**Figure 7**). So if you want to calculate the time, you cover 'time' and observe that distance appears above speed, so you calculate time from distance divided by speed. Similarly, to calculate speed, you cover 'speed' and observe that distance appears above time so you divide distance by time. For calculating distance, you cover 'distance' and note that time is alongside speed, so you multiply the two together.

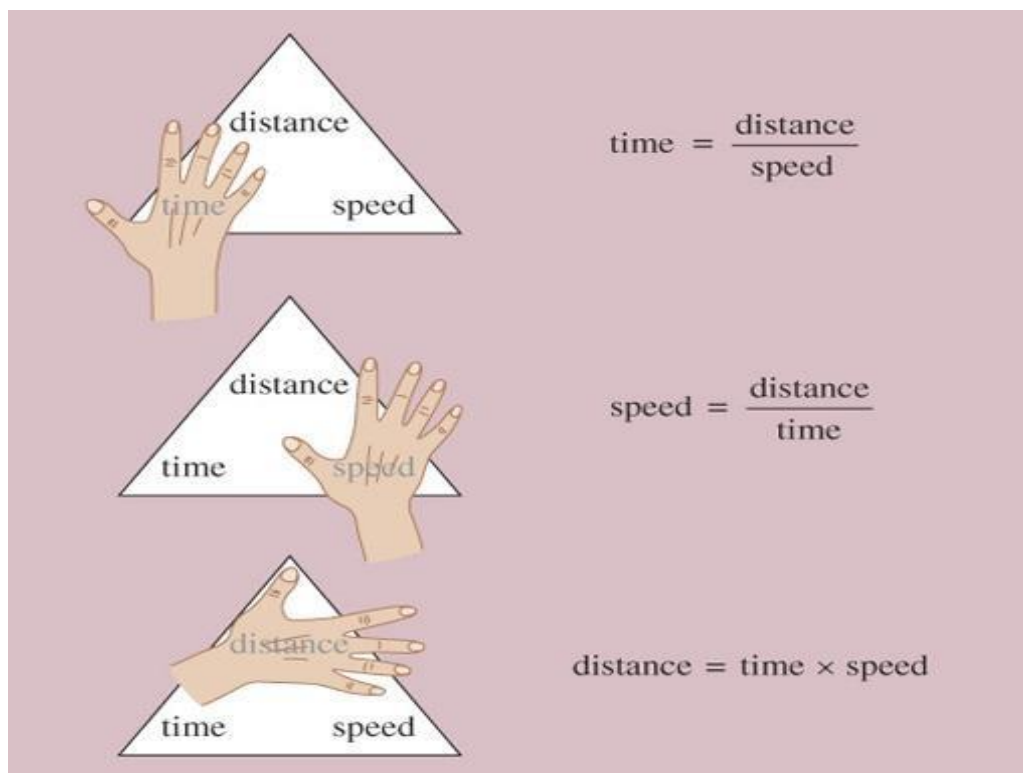
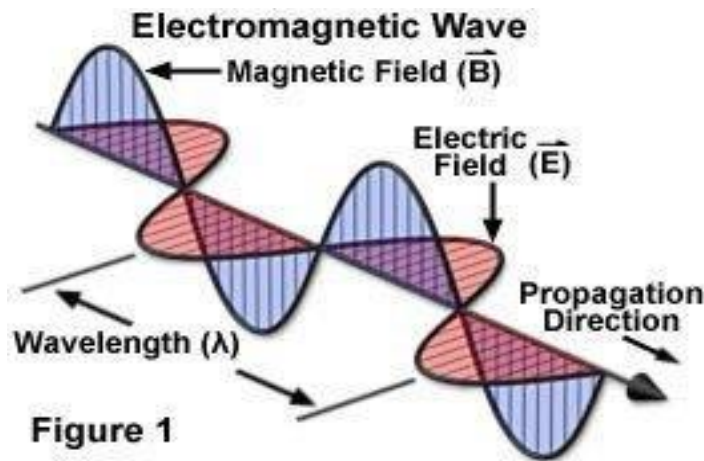


Figure 7 Using the formula triangle

Electromagnetic Signal Transmission

Electromagnetic signal

electrical wave and magnetic wave (bundled together with same frequency and amplitude) oscillating in mutually perpendicular direction, orthogonal to the direction of propagation of the signal. Does not need a medium for propagation. eg: light, heat, gravity etc



Electromagnetic waves

Electromagnetic waves were first postulated by James Clerk Maxwell and subsequently confirmed by Heinrich Hertz. Electromagnetic waves are created as a result of vibrations between an electric and a magnetic field. In this article, we will explore the definition and formation of electromagnetic waves along with the graphical and mathematical representations of electromagnetic waves in detail.

What Are Electromagnetic Waves?

Electromagnetic waves are also known as EM waves. Electromagnetic radiations are composed of electromagnetic waves that are produced when an electric field comes in contact with the magnetic field. It can also be said that electromagnetic waves are the composition of oscillating electric and magnetic fields. Electromagnetic waves are solutions of Maxwell's equations, which are the fundamental equations of electrodynamics.

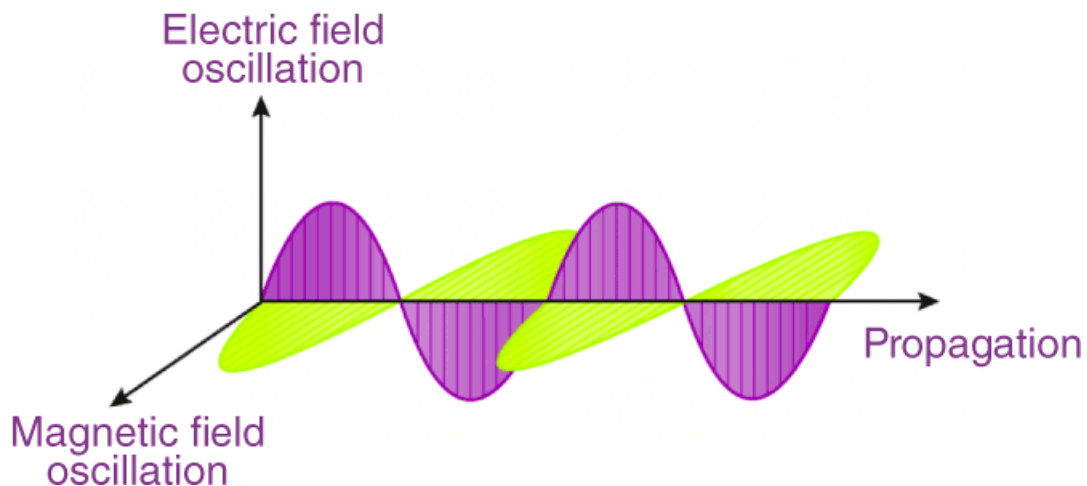
How Are Electromagnetic Waves Formed?

- Generally, an electric field is produced by a charged particle. A force is exerted by this electric field on other charged particles. Positive charges accelerate in the direction of the field and negative charges accelerate in a direction opposite to the direction of the field.
- The Magnetic field is produced by a moving charged particle. A force is exerted by this magnetic field on other moving particles. The force on these charges is always perpendicular to the direction of their velocity and therefore only changes the direction of the velocity, not the speed.

- So, the electromagnetic field is produced by an accelerating charged particle. Electromagnetic waves are nothing but electric and magnetic fields travelling through free space with the speed of light c . An accelerating charged particle is when the charged particle oscillates about an equilibrium position. If the frequency of oscillation of the charged particle is f , then it produces an electromagnetic wave with frequency f . The wavelength λ of this wave is given by $\lambda = c/f$. Electromagnetic waves transfer energy through space.

Graphical Representation of Electromagnetic Waves

ELECTROMAGNETIC WAVES



Electromagnetic waves are shown by a sinusoidal graph. It consists of time-varying electric and magnetic fields which are perpendicular to each other and are also perpendicular to the direction of propagation of waves. Electromagnetic waves are transverse in nature. The highest point of the wave is known as the crest while the lowest point is known as a trough. In vacuum, the waves travel at a constant velocity of $3 \times 10^8 \text{ m.s}^{-1}$.

Mathematical Representation of Electromagnetic Wave

A plane Electromagnetic wave travelling in the x -direction is of the form

$$E(x, t) = E_{max} \cos(kx - \omega t + \Phi)$$

$$B(x, t) = B_{max} \cos(kx - \omega t + \Phi)$$

In the electromagnetic wave, E is the electric field vector and B is the magnetic field vector.

Maxwell gave the basic idea of Electromagnetic radiations, while Hertz experimentally confirmed the existence of an electromagnetic wave.

The direction of propagation of the electromagnetic wave is given by the vector cross product of the electric field and magnetic field. It is given as:

$$\vec{E} \times \vec{B}$$

Electromagnetic Wave Equation

- The electromagnetic wave equation describes the propagation of electromagnetic waves in a vacuum or through a medium.
- The electromagnetic wave equation is a second-order partial differential equation.
- It is a 3D form of the wave equation.
- The homogeneous form of the equation is written as

$$\begin{aligned}(v_{ph}^2 \nabla^2 - \frac{\partial^2}{\partial t^2})E &= 0 \\ (v_{ph}^2 \nabla^2 - \frac{\partial^2}{\partial t^2})B &= 0\end{aligned}$$

Where,

$$v_{ph} = \frac{1}{\sqrt{\mu\epsilon}}$$

Intensity of an Electromagnetic Wave

$$\begin{aligned}I &= \frac{P}{A} = \frac{1}{2} c \epsilon_0 E_0^2 \\ &= \frac{1}{2} \frac{c}{\mu_0} B_0^2\end{aligned}$$

Speed of Electromagnetic Waves in Free Space

It is given by

$$C = \frac{1}{\sqrt{(\mu_0 \epsilon_0)}}$$

Where,

μ_0

is called absolute permeability. Its value is

$$1.257 \times 10^{-6} Tm A^{-1}$$

ϵ_0

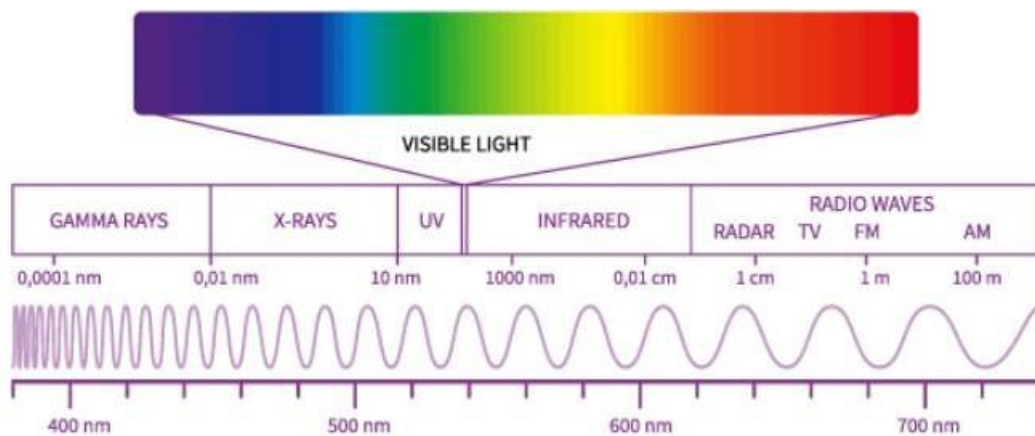
is called absolute permittivity. Its value is

$$8.854 \times 10^{-12} C^2 N^{-1} m^{-2}$$

C is the velocity of light in vacuum = velocity of electromagnetic waves in free space =

$$3 \times 10^8 m s^{-1}$$

Electromagnetic Spectrum



Electromagnetic waves are classified according to their frequency f or according to their wavelength

$$\lambda = \frac{c}{f}$$

The wavelength ranges of different lights are as follows,

For visible light – approx. 400 nm to approx. 700 nm

For violet light – approx. 400 nm

For red light – approx. 700 nm

Applications of Electromagnetic Waves

Following are a few applications of electromagnetic waves:

- Electromagnetic radiations can transmit energy in a vacuum or using no medium at all.
- Electromagnetic waves play an important role in communication technology.
- Electromagnetic waves are used in RADARS.
- UV rays are used to detect forged bank notes. Real banknotes don't turn fluorescent under UV light.
- Infrared radiation is used for night vision and is used in security cameras.